Policy Memo
for
Crop Residue Management and Utilization
in the states of Punjab and Haryana

Executive Summary

The memo delves into the several causes that entail the issue of crop residue management and the consequent stubble burning which range from the technicalities of the farming practices and farming seasons to gaps in the Punjab Preservation of Subsoil Water Act to the mechanisation methods. The paper attempts a status quo of Punjab and Haryana, with a comparative analysis of the states in their stances. The memo proposes two alternatives and offers an appropriate one after conducting a feasibility test.

Introduction

India is the **second largest** agro-based economy. Around **58%** of its population depends on **agriculture.** A year-round crop cultivation generates a large amount of agricultural waste, including crop residue. Paddy and wheat cropping is one of the major farming practices in the northwestern part of India, especially Punjab and Haryana. The northwestern states are famously known for their contribution of approximately **26% and 59%** to the country's total **rice and wheat production**, respectively. However, they are also equally infamous for their post-harvest Crop Residue Burning Practices. The two states **Punjab and Haryana** alone produce **48% of the whole straw production in India**, which is **openly burned in situ**. Improper disposal of crop residue in India is a major factor behind rising carbon emissions, soil degradation, air pollution, emission of toxic fumes, and so on. The states that initially

benefited from the Green Revolution in India are now facing significant problems because of it. Over time, farmers in Punjab, Haryana, have transitioned to a specialized approach, focusing on high-yielding rice-wheat cultivation, which is considered efficient but has incurred substantial environmental and health repercussions. Despite the prohibition imposed by the Punjab Pollution Control Board and The National Green Tribunal (NGT) the detrimental practice of stubble burning persists. **Stubble burning entails the deliberate setting of fire to the leftover plant material once the valuable parts of the crop have been harvested.**

Crop residue burning adds significant emissions load to our already existing choked cities. **When 1 tonne of crop residue is burnt on field, it releases approximately 1400kg of ,58 CO2 kg of CO,11kg of particulate matter (PM),4.9kg of NOx and 1.2 kg of SO2.**Due to such high amount of emissions crop residue burning has become a major problem in India. A report from the **System of Air Quality and Weather Forecasting and Research (SAFAR)** found that in 2019, crop burning made up about **10-20% of the air pollution in Delhi** and nearby areas during October and November. On some days when a lot of burning happened, it was as high as **44%.** During this time, the **air in Delhi was 12-20 times worse** than what the **World Health Organization says is safe**. Not only does this burning create more pollution and smog, but it also **hurts nature by reducing biodiversity, harming farmland, and making the soil less fertile.**

The Government of India as well as the State Government of Punjab and Haryana have already come up with a policy intervention in this domain. There is a Central Sector Scheme on ***'Promotion of Agricultural Mechanization for In-Situ Management of Crop Residue in the States of Punjab, Haryana, Uttar Pradesh and NCT of Delhi.'*** Under this financial

assistance @ 50% of the cost of machinery is given to the farmers for purchase and @80% of the project cost is provided to Cooperative Societies of Farmers, Farmer's Producers Organizations (FPOs), Registered Farmer's Societies, and Panchayats for the establishment of Custom Hiring Centres of crop residue management machinery. However, we have identified substantial gaps in the framework and the implementation of this policy. **The policy problem which we have identified is the lack of targeting the sources of pollution at ground levels and streamlined framework that necessitates implementation to ensure extensive adoption of other alternatives of stubble burning in India.**

Methods

To better examine the issue, we conducted a thorough study using both qualitative and quantitative research methods. This research formed the basis of our understanding regarding the necessity of taking action. Our study drew upon various sources, including:

● Government issued guidelines on crop residue management and stubble burning: The guidelines provided us with information regarding the limitations of the current measure and also helped us understand the various limitations in the existing policies.

● Academic sources, laboratory reports and field tests, data records by

intergovernmental and international organisations: These sources helped us learn more about the effects of stubble burning on the environment and human health. They also expanded our research by presenting different alternative approaches to mitigate the problems caused by stubble burning, along with their respective drawbacks.

● **Punjab Agricultural University Report**, "*Impact Assessment Study of Central Sector Scheme for Promotion of Agricultural Mechanization for In-situ Paddy Stubble Management*

in Punjab" released in 2022 The report helped us in understanding the shortcomings of the Central Sector Scheme for Promotion of Agricultural Mechanisation for In-situ Paddy Management. It further provided various strategies that could help in scaling up the adoption of Paddy Stubble Management.

● Newspaper articles reporting the implementation: It gave insight into the on-ground situation of stubble burning in India.

Issue Analysis

Our research pointed us to several factors that entail the menace of subtle burning. For instance, the subject of **low nutritional value of straws**, specifically rice straws. Punjab and Haryana produce coarse rice which are very different from Basmati. The difference ranges from texture to nutritional value. Unlike basmati straws, which are used for feeding the livestock because of a better nutritional composition, the coarse rice straws are not utilised the same. More rice straws are burnt in comparison to wheat due to more production of coarse rice. In addition there exist **outdated modes of utilising the stubble**. Earlier the stubble was used by farmers as fuel for cooking, raw material for roof, fodder for livestock, etc. Many of these solutions became outdated with time and even after implementation, leave a large portion of stubble. In addition to this, there has been concern around the **scarcity and cost of labour**. After MGNEREGA (2005) was introduced, a sudden shift was observed in the occupation of labours. Many labours who used to come from Bihar and UP to work made a shift in their jobs after the implementation of the scheme. This created a shortage of labour. A rise in the prices and low labour population moved the farmer's choice from a labour intensive to a capital-intensive mode of harvesting. Moreover, the **use of machines (combine harvesters)** poses new challenges. Before machines, stubble was removed manually leaving

no residue behind. But after an increase in the area of harvesting-making the manual labour a very time-consuming process, rise in the labour prices, accessibility of machines, many farmers rejected the traditional method and went for a more capital-intensive process of harvesting. However, because of the inability of the machines to remove the straws entirely, spreading it all over the field, many sought to burn it.

However, a key matter of concern among these is the subject of **cropping patterns**. Green revolution which transformed India agriculture, led to some unfavourable consequences. The new HYV varieties of wheat and rice have led to increased irrigation and fertiliser application. Also, the mechanization of agriculture, such as the use of combine harvesters, leaves behind smaller stubble on the ground that is difficult to collect. These varieties have also marked a shift in the cropping patterns. For instance, in Haryana, during 1965-66, the dominant food crop was bajra (46%). However, by 1995-96, rice (34%) was the major crop[1]. Similar was the case for Punjab, where area under rice cultivation increased from 6.05% in 1960-61 to 63.02% in 2004-05[2]. Apart from altering dietary consumption patterns, this shift has also led to crop residue burning. Since the hybrid crops have crop cycles of shorter duration, instead of the traditional crop cycles, the farmers are left with no time but to prepare the land by burning the agricultural waste.

Moreover, crops such as rice are water-intensive leading to water stress in various parts. India has the highest demand for freshwater usage globally, and 91% of water is used in the agricultural sector now (Kayatz et al., 2019).[3] Thus there have been efforts to conserve groundwater. Laws like the **Punjab Preservation of Subsoil Water Act** delays the date for paddy transplantation to beyond June 10th, when the most severe phase of evapotranspiration

[1] Singh 2000
[2] Toor et al cited in Lakhwinder Singh (2013)
[3] Kayatz, B., Harris, F., Hillier, J., Adhya, T., Dalin, C., Nayak, D., et al. (2019).

(transfer of water from land to the atmosphere through evaporation from the soil and plant transpiration) is over. Farmers are forbidden from sowing paddy before May 10th, and transplanting it before June 10th. Farmers' organisations say late sowing and transplanting delays the harvesting as well (it is end-October by the time operations end), and they are left with a very small window to prepare their fields for the next (wheat) crop. In this situation, setting the stubble ablaze is a quick-fix solution.

Status Quo

Current Action Plan of Punjab: As we all know; Punjab's major crops are rice and wheat; which are the major crops that cause stubble burning. To address this issue, the government of Punjab has come out with a notice in 2022 namely "Action Plan for Control of Burning of Crop Residue in the State of Punjab" which is still being widely followed. The Action Plan calls for various Central and State incentivization programs to stop stubble burning. Various government officials have been directed to take severe action against stubble burners under section 19(5) of the Air (Prevention and Control of Pollution) Act, 1981. Many initiatives by the State and the Central Departments to eradicate stubble burning. A few of them are: -

- Implementation of "**Central Sector Scheme for Promotion of Agricultural Mechanization for In-situ Management of Crop Residue**" which provides a 50% subsidy for individual farmers and 80% for groups and cooperative societies to purchase machinery such as CRM (Crop Residue Management) Machines and Happy Seeder.
- The State Department is collecting the paddy straw to use as cattle feed for landless farmers within the State and outside the State. The paddy straw is also used in other industries like paper and cardboard mills.

- Crop Rotation practices have been proven to be economically viable and acceptable as they reduce the negative externality on the environment and are profitable for the farmers. A few examples are cotton-wheat, maize-wheat, etc.
- The Ministry of Agriculture provides Paddy Straw Management Compensation to the Farmers at rupees 100 per quintal of paddy straw produced for at least 3 years.

There have been various instructions that have been passed down to the District Magistrates and other district-level officers by the Punjab Pollution Board to penalize the farmers who practice Stubble Burning and to ensure the mandatory use of the Super SMS with combined Harvester. Other forms of measure such as door-to-door awareness programs, display boards, print media advertisements, and pamphlet distribution to the farmers have opted to create general awareness about the ill effects of Stubble Burning in terms of environmental and economic hazards.

Current Action Plan of Haryana: Haryana has been a part of the initial green revolution which subsequently led to large-scale wheat and paddy cultivation. With even increasing demand for staple foods, the issue of stubble burning has also become a matter of concern. The State of Haryana has put the Commission for Air Quality Management (CAQM) on board to provide directives to the Chief Minister's Office (CMO) and the District Collectors (DC) directly to ensure effective and timely implementation of the Action Plan to arrest Stubble Burning. The State of Haryana has been divided into three zones namely - Red, Yellow, and Green; in descending order in terms of the Active Fire Location (AFL), and disburse the State apparatus accordingly. Many financial incentive programs have been launched to incentivize the farmers not to burn the Paddy Straw. A few of them are: -

- Rs. 1000 /- per acre for in-situ/ ex-situ management of paddy crop residue.
- Rs. 7000/- per acre for diversification of paddy area with alternative crops under the Mera Pani Meri Virasat scheme.
- Rs. 4000/ - per acre for adoption of direct seeding of rice.

- Rs. 1,00,000/- for achieving Zero Burning to Red Zone Panchayats.
- Transportation charges of Bales @ Rs 500/ - per acre are limited to a max of Rs. 15,000/- to Gaushalas.
- Fixation of rates for procurement of paddy crop residue @ Rs. 2,500 per tonne.
- Special provisions of providing subsidies to the clusters identified for 2G ethanol plant

The State has well over 80,000 crop residue management machinery that is not directly linked to the farmers and the incentivization scheme has been much more viable as the farmers now have a financial incentive not to burn the paddy straw. The State government is collecting the residue as immediately as possible to ensure that the farmers can keep their farms ready before the next crop. The use of Drone technology and satellite mapping has helped survey the real-time effect of stubble burning and implementing speedy solutions.

Comparative Analysis between both States: The State of Punjab had about 60 lakh hectares of land under Paddy Cultivation with about 18.74 million tonnes of Paddy Straw Generated. For Haryana, the total area under paddy is estimated to be around 14.82 Lakh Hectares and generates about 10 million tonnes of Paddy Straw. This clearly shows that Haryana has a higher generation of Paddy Straw but produces a lesser amount of Smog in a comparison to its neighbouring state due to the financial incentive programs inducted by Haryana. Even though this acts as a short-term solution for this complex problem, it gives the State and the Central Government enough time to formulate a long-term solution for this issue. And the cost that is to be borne must be distributed between the Centre and the State which equates it to the social cost that the affected states have been bearing. This plan has been adopted by Haryana which yielded viable results. The reduction in the chain of command by Haryana by directly involving the Commission for Air Quality Management (CAQM) with the Chief Minister’s Office (CMO) and the District Collectors (DC) led

to the understanding of the issue much better as the authorities are closer to the scene of action and the reporting is done directly to the CMO.

Proposed Solutions

Alternatives and changes in the Status-Quo: Despite the formulation of the existing policies, it is necessary to understand the feasibility of these policies in the current scenario and consider alternatives to ensure that the mitigation of the problem is fully done. Stubble Burning is still widely practiced in India due to the vast amounts of Wheat and Paddy plantation in the North-western belt of India. It involves setting fire to the straw stubble that remains after grains, such as rice and wheat, have been harvested. Stubble Burning is not only restricted to India but is a pestering problem across many countries such as the United States of America, Brazil, China, Argentina, Mexico, and many more. Thus, it is crucial to evaluate their policies to get a better understanding of the issue and formulate the best possible policy. Mentioned below are two alternatives to address the issue with lesser cost and better efficiency.

Alternative 1: The Stubbles can be handled by the process of the accelerated decomposition method proposed by the Indian Agriculture Research Institute (IARI) with the use of a microbial cocktail known as “Pusa Decomposer”. This process takes around 25 days to completely decompose the stubbles on the farm within 25 days and will cost less than Rs.1000/- per acre. If we shift to short-duration varieties like- Pusa Basmati 1509 and PR-126 in the long-term, these variants shall widen the window between rice harvesting and wheat sowing; which gives enough time for the “Pusa Decomposer” to completely decompose the Stubble. The fundamental drawback of this method is due to the amount of time it consumes. Most of the Farmers are not willing to wait for 20-25 days after the harvest is done and thus, they find Stubble Burning as an easier alternative to

do, for this method to roll out efficiently requires large-scale social awareness and stringent rules to implement.

Alternative 2: To roll out an immediate plan to arrest the stubble-burning issue, the use method of "Mulch and Happy Seeder" seems as a viable option. It entails a two-fold process of straw and stubble management and seeding. A layer of undecomposed material is applied to the surface of the soil and then the Happy Seeder shreds the straws and seeds wheat simultaneously. With the help of large-scale government subsidies and awareness amongst the farmers, over 5,000 hectares of land were using the Happy Seeder and the projected growth of this method is estimated to be around 700,000 by 2025. This alternative helps in the improvement of the soil condition, moisture, and texture and reduces weed growth. However, the fallback in this method is the high cost of the tractor and the ongoing variable cost to operate the Happy Seeder. It has one of the lowest CO_2 outputs about 0.93 tonnes per hectare.

The Feasibility Test

Alternatives ⇩ Criteria ⇨	Economic Efficiency	Environmental Efficiency	Financial Feasibility	Political Feasibility	Operational Feasibility	Societal Benefit
Status Quo	✓		✓		✓	
Alternative 1	✓	✓	✓	✓	✓	✓
Alternative 2		✓		✓		✓

The above feasibility test clearly shows us the various criteria that are to be taken into consideration and how the “PUSA” Decomposer is a much more viable option in terms of all possible criteria as it is even easier to implement in terms of administration. The cost efficacy and efficiency in terms of implementation for the farmers make it the most viable option.

Recommendations

After analyzing that the Status Quo is not a viable option as it is highly environmentally hazardous and results in the ever-increasing pollution of the nation. This is an economically viable option for the farmers but the social cost borne by the nation far outweighs the benefit of the agriculturists. The second alternative i.e., “Mulch and Happy Seeder” is much better compared to the Status Quo but has very little economic efficiency. The Happy Seeder machine costs about 1.75 lakhs to 2.5 lakhs. This is a huge upfront cost’ especially for the marginal farmers. The government of Punjab already tried to work a model for the state by providing subsidies for the farmers and renting out the Happy Seeder but still, it costs up to 2000 rupees as a part of the operational cost thus making it unviable to small landowners. The “PUSA Decomposer” is the most viable option to go ahead with. It covers all ambits of economic viability, Political Feasibility, Financial feasibility, and many more. After decomposing the stubble, it saves the farmer from using any further fertilizer as the stubble acts as a natural manure for the farmland. This method is the best possible solution to utilize and implement.

Action Plan to implement “Pusa Decomposer”

The “Pusa Decomposer” is a consortium of Fungi that was developed by the Indian Council of Agricultural Research (ICAR) to tackle the issue of Stubble Burning. The process of using the Pusa Decomposer is to mix 25 liters of liquid with 4 capsules of the decomposer with jaggery and Besan

and it suffices for about 1 hectare of land. The fermentation process takes up to 8-20 days and it takes about another 25 days to fully decompose the Stubble. The respective State Governments must heavily advocate the economic benefits of the Pusa Decomposer and if necessary, must provide them subsidy in obtaining the tablets for the Pusa Decomposer. There must be a proper survey of all the districts in the State and identify the major zones of impact of Stuble Burning and later direct the District Magistrates (DM) to implement this method effectively in those districts. There must also be a team of Technicians to guide the farmers on how to use this composition and this must also be relayed on the official YouTube channels and the News Channels regularly for the convenience of the farmers. The benefits of the “Pusa Decomposer” are to be discussed at the ground level such as the Panchayat Raj and Block Samithis to create general awareness amongst many villagers. Since this method has already been heavily implemented in a few districts across Haryana, Punjab, and Uttar Pradesh and the results have been fruitful, it is easier to convince the public of the possible benefits for the farmers.

Limitations

There are a few fundamental drawbacks to the use of the “Pusa Decomposer.” It takes almost 35-40 days to ferment and apply the decomposer in the land which leaves a very short window for the farmers to sow their next crop which somewhat makes it unviable for farmers who heavily depend on agriculture as a source of income and wish to sow at least 3 crops a year. The capsule is inherently used for non-basmati rice and this becomes a problem for Paddy and wheat farmers; though a variant for these crops is also being developed. The Pusa decomposer also faces an issue when the moisture content in the stubble varies from the usual resulting in a longer time to decompose and delaying the next sown crop. The temperature conditions for the “Pusa Decomposer” to work at its best is between 32° C to 36°C, and, any sort of anomaly in the temperature leads to a delay in the timeframe or a lack of efficiency in the “Pusa Decomposer”. Thus, all these plethora of issues are to be addressed by the Central and state governments in

collaboration with scientific bodies such as ICMR, ICAR, etc, and formulate a short-term and long-term policy for complete mitigation of Stubble burning and subsequently incentivize the farmers, if possible.

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